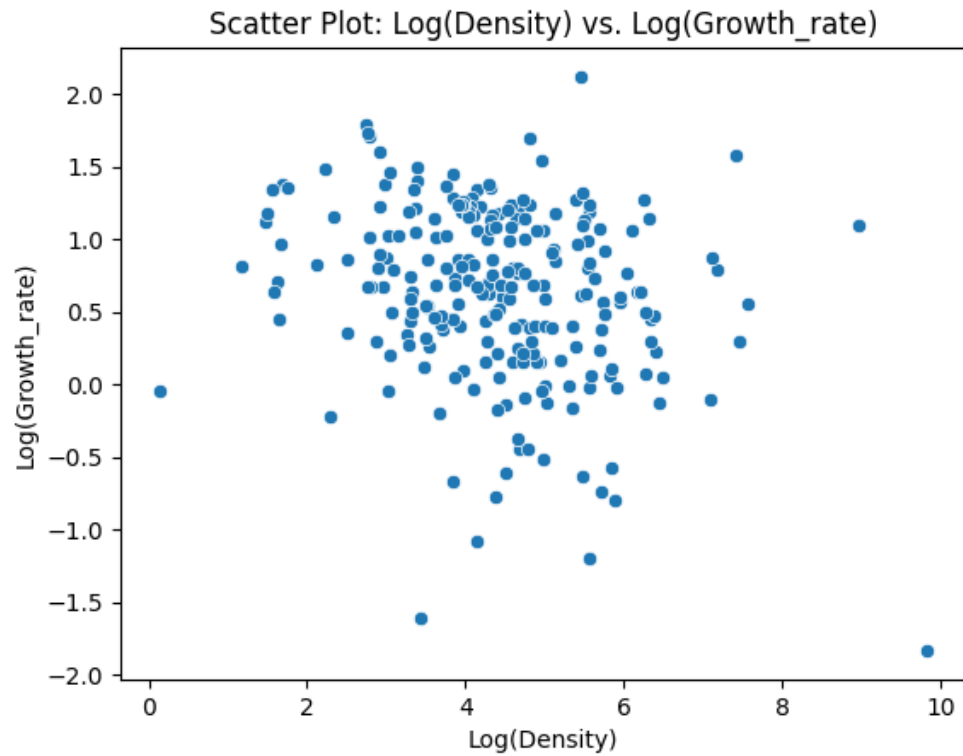


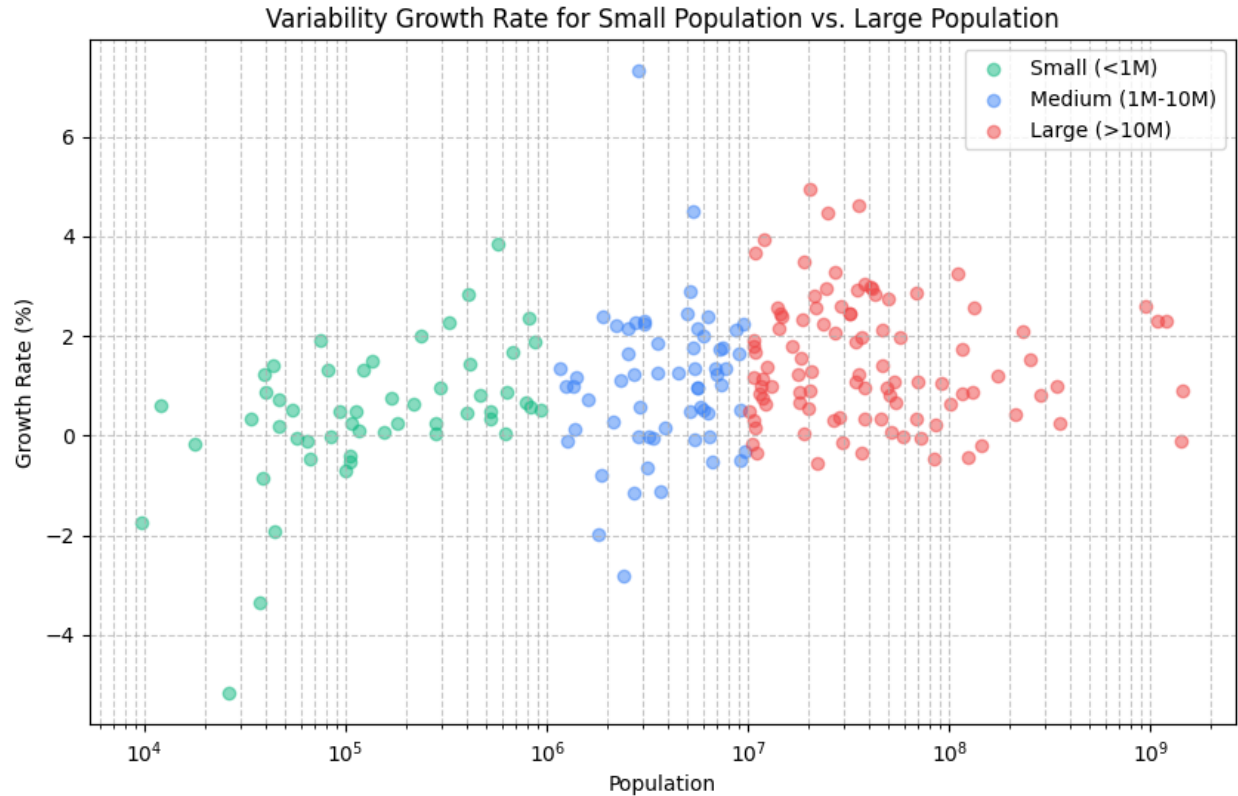
VISUALIZATIONS FOR HYPOTHESIS TESTING

HYPOTHESIS 1



The figure shows a scatter plot between the Log (Density) against Log (Growth Rate) to determine the correlation between the two features. The visual shows a weak correlation between density and growth rate. The weak correlation could be attributed to factors such as economic conditions, policies or regional factors as the primary causes for growth rate rather than population density.

HYPOTHESIS 2:

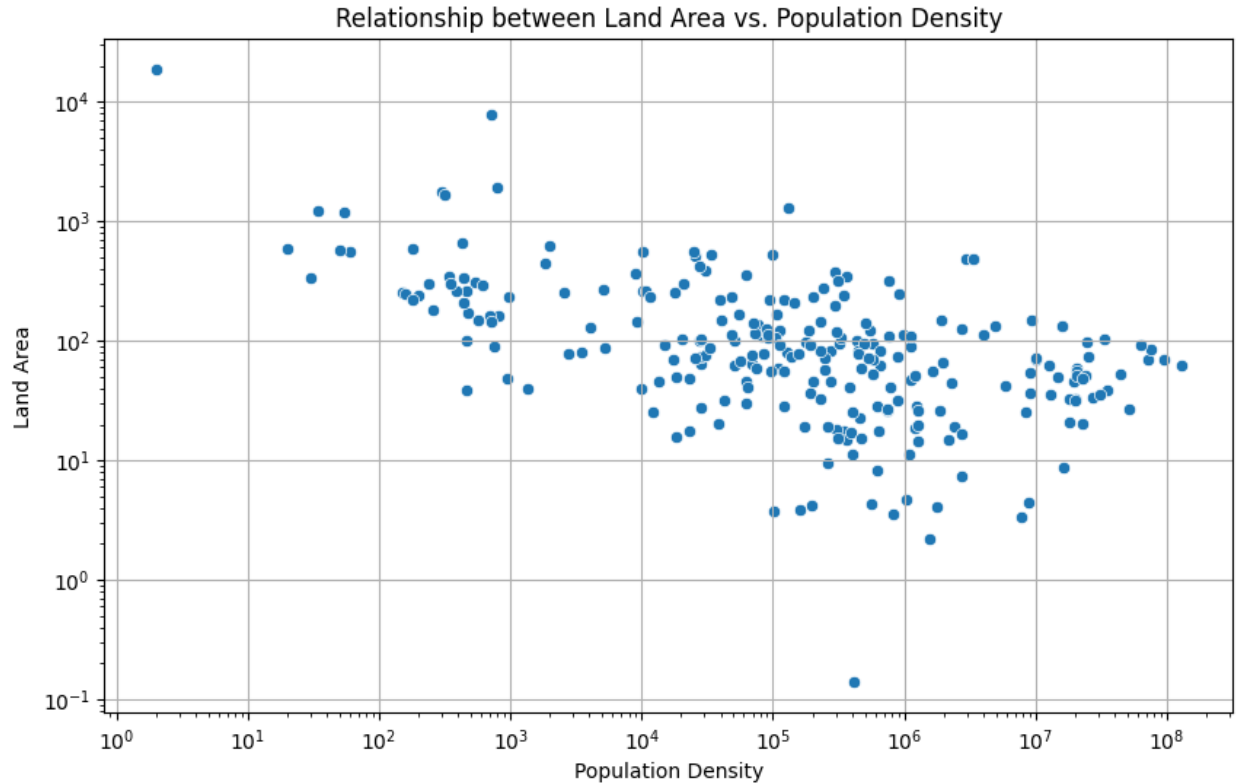


The visual shows a plot to determine the relationship between population and growth across different population sizes, specifically between small countries and large countries. The plot indicates that countries with small populations show increased variability. This could be a result of external factors like migration or economic shift. The large population countries show a stable growth pattern because of a stable economic trend.

HYPOTHESIS 3:

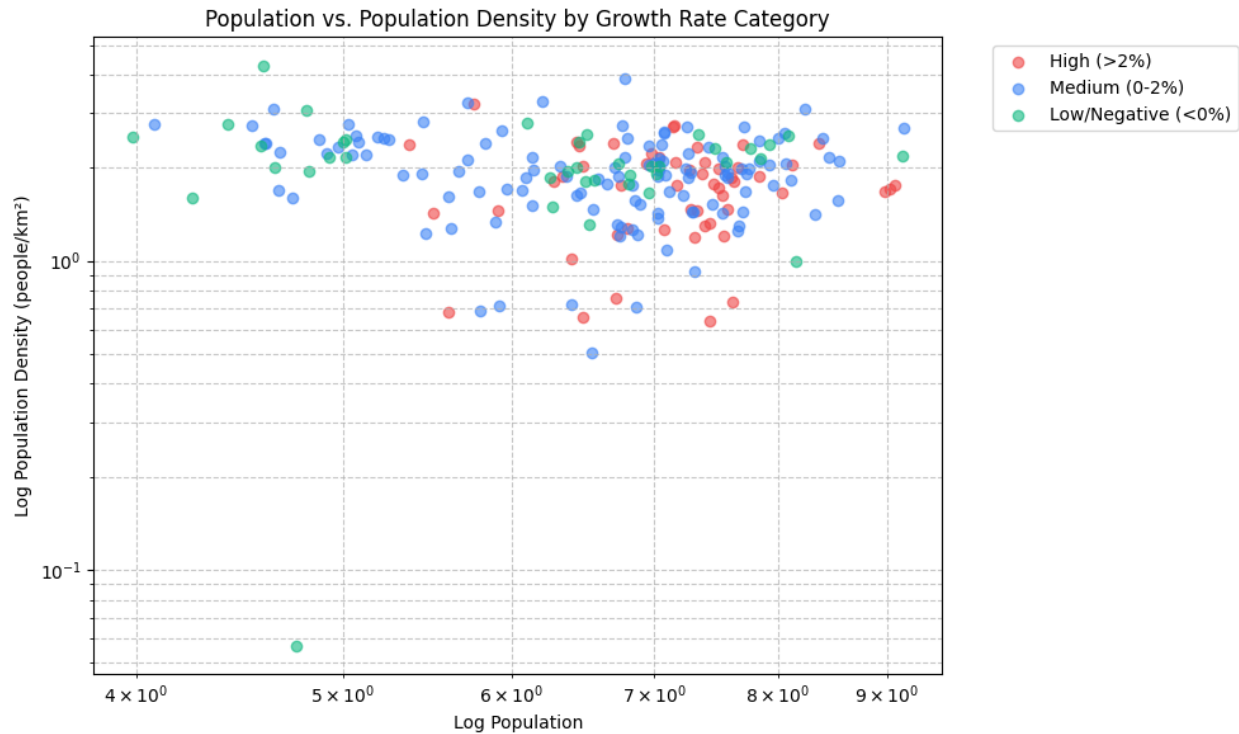
A suitable correlation plot was not possible to determine the relationship between Negative growth rate and regional patterns because not many countries with negative growth rates are available in the data.

HYPOTHESIS 4:



The plot shows a very weak negative correlation with no meaningful relationship between Land area and Population density. The plot shows an increasing downward trend that shows a negative relationship between land area and population density. The inverse relationship suggests that population density is influenced by land availability, but the wide scatter indicates that other factors (e.g., economic development, urbanization, or climate) also play significant roles.

HYPOTHESIS 5:



The scatter plot establishes a relationship between population and population density by growth rate category. There is no strong linear trend between population and population density across all categories. However, the nature of the clustering of medium-growth (blue) points suggests that the highly populated countries maintain stable growth rates, while high-growth (red) and low/negative-growth (green) points show more variability, possibly due to economic, policy, or regional factors. The presence of high-growth countries at both low and high densities aligns with the hypothesis.